

Tuong Luu

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EDUCATION

University of Waterloo

Candidate for Bachelor of Honours Computational Mathematics

Waterloo, Ontario

Sep 2023 - May 2028

TECHNICAL SKILLS

Languages: Java, Python, FastAPI, C/C++, JavaScript/TypeScript, Node.js, Express, React, Next.js

Databases: PostgreSQL, MongoDB, GCP (GKE, Secret Manager)

Tools: Git, Docker, Kubernetes (K8s), Linux, Bash, VS Code, Postman

Certifications: Azure Fundamentals, Azure AI Fundamentals

EXPERIENCE

Software Engineer Intern

Carta – Incoming Summer 2026

Kitchener, Ontario

May. 2026 – Present

Software Developer Intern

Lynkr Inc

Toronto, Ontario

Sep. 2025 – Dec. 2025

- Collaborated with engineers to develop a FastAPI and Next.js application with AI agent orchestration, generating **\$100K+ CAD** in revenue and supporting **500+ beta users**
- Migrated the database seeder to a Kubernetes job, reducing deployment time on GKE by **40%**
- Architected multi-organization support with tenant isolation and RBAC, enabling secure segregation for and independent configuration per tenant
- Built account security features with MFA using TOTP and Twilio, along with JWT session controls, cutting unauthorized access by **70%** and protecting **500+** active sessions
- Developed an audit logging system tracking all user actions, admin changes, and security events, supporting compliance for **100% of critical operations**

WE Accelerate Program - Azure and Artificial Intelligence

University of Waterloo

Waterloo, Ontario

Jan. 2025 – Apr. 2025

- Led a team of 5 to design an AI-based solution preventing financial fraud against elders.
- Developed a comprehensive project planning tool and pipeline sequence wireframe, visualizing each step and aligning deliverables with industry best practices.
- Collaborated closely with a project mentor to refine goals, validate assumptions, and incorporate **real-world AI considerations** into the solution.

PROJECTS

AniDis - Anime Discussion Platform | React, TypeScript, Node.js, Express, MongoDB, Docker

- Developed **AniDis**, a MERN-based anime discussion platform with real-time updates for **1,000+** titles and threaded comments supporting unlimited nested replies.
- Automated data ingestion and trend updates with scheduled backend jobs, ensuring daily content freshness and scalable performance.
- Containerized deployment with Docker/Docker Compose, environment-based configuration, and CI/CD-ready Git workflows; self-hosted with Cloudflare Tunnel.
- Implemented security best practices and optimized frontend rendering for high-performance discussions.

Kyzen - 2D Game Engine | Java, LWJGL, Maven, OpenGL

- Developed a 2D game engine with batch rendering, optimizing the rendering of multiple objects in a single batch for improved performance by up to **60%** when handling **1000+** objects per frame
- Utilized OOP principles to create scalable software, implementing design patterns like Builder and Singleton to ensure modularity and maintainability
- Implemented an Entity-Component System (ECS), providing a modular and extendable architecture for flexible game object composition
- Integrated texture atlas support, enabling efficient texture management for sprites and tiles, reducing draw calls by over **70%**